

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION**

ORDER NO. 99-081

**WASTE DISCHARGE REQUIREMENTS
FOR
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY**

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The U. S. Army Corps of Engineers (hereafter Discharger) owns and operates a campground at Lake Success, approximately two miles northeast of Success Dam and seven miles east of Porterville.
2. Discharge at the facility is currently governed by Waste Discharge Requirements Order No. 86-110, adopted by the Board on 30 May 1986. Order No. 86-110 is neither adequate nor consistent with current plans and policies of the Board.
3. The campground has 104-unit spaces, seven restrooms, and a recreational vehicle pump out station. Wastewater is treated in five septic tanks. Effluent from the tanks is transported through a force main to two disposal ponds.
4. The disposal ponds occupy a total area of one acre. Only one pond is regularly used. The other pond serves as a backup and has not been used since its construction.
5. The treatment and disposal system was designed for a flow of 12,000 gallons per day. Current flows in summer range only between 600 to 650 gallons per day and winter flows range from 200 to 250 gallons per day.
6. Source water for the campground is supplied by an on-site well.
7. Soils at the disposal area consist of San Joaquin Loam. The surface layer of this soil is typically brown to reddish brown loam, about 13 inches thick. The subsoil is reddish brown sandy clay loam and clay about 12 inches thick. Below the subsoil is a yellowish red hardpan about 31 inches thick. Below the hardpan is brown, stratified sandy loam and loam to a depth of 78 inches. This San Joaquin soil has very slow permeability in the subsoil and in the hardpan.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-2-

8. The campground is in Sections 24 and 25, T21S, R28E, MDB&M, as shown in Attachment A, which is part of this Order by reference. Surface water drainage is to Lake Success. The site lies within the Springville Hydrologic Subarea of the Tule River Hydrologic Area (No. 555.12) in the Southern Sierra Hydrologic Unit, as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
9. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the Basin. These requirements implement the Basin Plan.
10. The beneficial uses of Lake Success, as identified in the Basin Plan, include agricultural supply, hydropower generation, water contact and non-contact water recreation, warm fresh water habitat, wildlife habitat, and freshwater replenishment.
11. The beneficial uses of groundwater in the area are domestic, industrial, and agricultural supply.
12. Based on the information obtained from the Department of Water Resources, the average annual rainfall in the area is 11 inches and annual average pan evaporation rate is 82 inches.
13. The Regional Board has considered antidegradation pursuant to State Board Resolution No. 68-16 and finds that the permitted discharge is consistent with those provisions, and is unlikely to cause an increase in groundwater constituents above that of background levels.
14. The action to update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, Section 15301.
15. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
16. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

**WASTE DISCHARGE REQUIREMENTS ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY**

-3-

IT IS HEREBY ORDERED that Order No. 86-110 is rescinded and U.S. Army Corps of Engineers, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at Success Reservoir Campground:

A. Discharge Prohibitions

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of treated or untreated waste is prohibited.
3. Discharge of waste classified as 'hazardous,' as defined in Section 2521 (a) of Title 23, CCR, Section 2510, et seq., or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications

1. The monthly average discharge shall not exceed 0.012 mgd.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the disposal area.
3. As a means of discerning compliance with Discharge Specification B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in the ponds shall not be less than 1.0 mg/l.
4. The maximum conductivity (specific electrical conductance at 25° C, also EC) of the discharge shall not exceed the source water EC plus 500 μ mhos/cm.
5. The septic tanks shall be operated and maintained in manner that precludes discharge of sludge and scum to the ponds. Specifically, the septic tanks shall either be inspected or pumped annually. If inspected, then the tanks shall be pumped whenever it appears that sludge or scum will begin to carry-over into the disposal ponds; when solids accumulation in any of the tank is 75% of the usable solids storage volume of the tank; the scum layer is within three inches of the bottom of outlet pipe; or the sludge level is within eight inches of the bottom of the outlet pipe, whichever is more restrictive.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-4-

6. Ponds shall not have a pH less than 6.5 or greater than 8.5.
7. Ponds shall be managed to prevent breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
8. Public contact with wastewater shall be precluded through such means as fences and signs, or acceptable alternatives.
9. The ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically to the lowest point of potential overflow).
10. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.9.
11. Solids, scum, grease and other wastewater materials shall not be visible on the surface of the ponds.

C. Sludge Disposal Specifications

1. Collected sludges, scums, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer and USEPA Regional Administrator at least **90 days in advance** of the change.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-5-

D. Groundwater Limitations

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except conductivity. Regarding conductivity, the Discharger shall not cause underlying groundwater to exceed an incremental increase in conductivity greater than 30 $\mu\text{mhos/cm}$ over any five-year period.

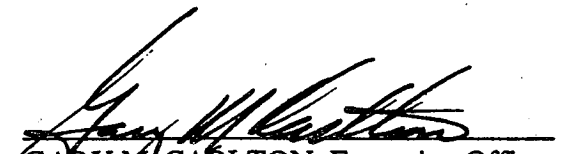
E. Provisions

1. The Discharger shall comply with Monitoring and Reporting Program No. 99-081, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. Septic tank cleanings shall be performed only by a duly authorized service.
4. The Discharger shall ensure that wastes from recreational vehicles (with constituents including formaldehyde, zinc, and phenol) do not deleteriously affect the septic system or impact groundwater.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. A copy of this Order shall be kept at the facility for reference by the personnel responsible for wastewater management. These persons shall be familiar with its contents.
7. The Board will review this Order periodically and will revise requirements when necessary.

WASTE DISCHARGE REQUIREMENTS ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-6-

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 11 June 1999.


GARY M. CARLTON, Executive Officer

SH:fmc:6/11/99

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 99-081

FOR
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to the Board and attached to this Program.

SEPTIC TANK MONITORING

All septic tanks shall be inspected and pumped as described below. An inspection is not required during the year a septic tank is pumped.

<u>Constituent</u>	<u>Units</u>	<u>Type of Measurement</u>	<u>Frequency</u>
Sludge Depth	feet	Staff Gauge	Annually ¹
Total Volume of Sludge (A)	Ft ³	Calculated	Annually ¹
Total Useable Sludge Storage in Tank (B)	Ft ³	Calculated	Annually ¹
Percent Full (A/B)	%	Calculated	Annually ¹
Scum Thickness	feet	Staff Gauge	Annually ¹
Distance Between Bottom of Scum Layer and Bottom of Outlet Device	inches	Staff Gauge	Annually ¹
Distance Between Top of Sludge Layer and Bottom of Outlet Device	inches	Staff Gauge	Annually ¹

-
- 1 In lieu of septic tank measurement, the septic tanks may be pumped annually. The Discharger shall maintain a log of all septic tank cleanings. At a minimum, the log shall include the date of the cleaning, and the name, address, phone number, and license number of the cleaner.

MONITORING AND REPORTING PROGRAM NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-2-

EFFLUENT MONITORING

Monitoring of septic tank effluent shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
Flow ¹	gallons/day	Estimate	Monthly
EC	µmhos/cm	Grab	Monthly
Formaldehyde	mg/l	Grab	Annually
Zinc	mg/l	Grab	Annually
Phenol	mg/l	Grab	Annually
Total Nitrogen	mg/l	Grab	Annually

¹ Flow estimates may be obtained from water meter readings and the Discharger shall maintain a separate log of flow from recreational vehicles.

POND MONITORING

The freeboard shall be monitored on all storage ponds to the nearest tenth of a foot. Monitoring of storage ponds shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
pH	pH Units	Grab	Monthly
Freeboard	feet	Observation	Monthly
Dissolved Oxygen ¹	mg/l	Grab	Monthly

¹ Samples shall be collected from opposite of the inlet to each pond and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

A permanent marker shall be placed in each pond with calibration including the water level at design capacity and available operational freeboard.

MONITORING AND REPORTING PROGRAM NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-3-

In addition, the Discharger shall inspect the condition of the ponds once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the ponds (e.g., dark sparkling green, dull green, yellow, gray, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the operation and maintenance manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action was taken or is scheduled to be taken.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency</u>
EC	µmhos/cm	Grab	Annually

REPORTING

Monthly monitoring reports shall be submitted to the Board by the **20th day of the following month**. Annual monitoring reports shall be submitted by **20 January of each year**.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss any corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

MONITORING AND REPORTING PROGRAM NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

-4-

By **20 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, certificate grades, and general responsibilities of all persons in charge of wastewater treatment and disposal.
- b. The names and telephone numbers of persons to contact regarding wastewater disposal for emergency and routine situations.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

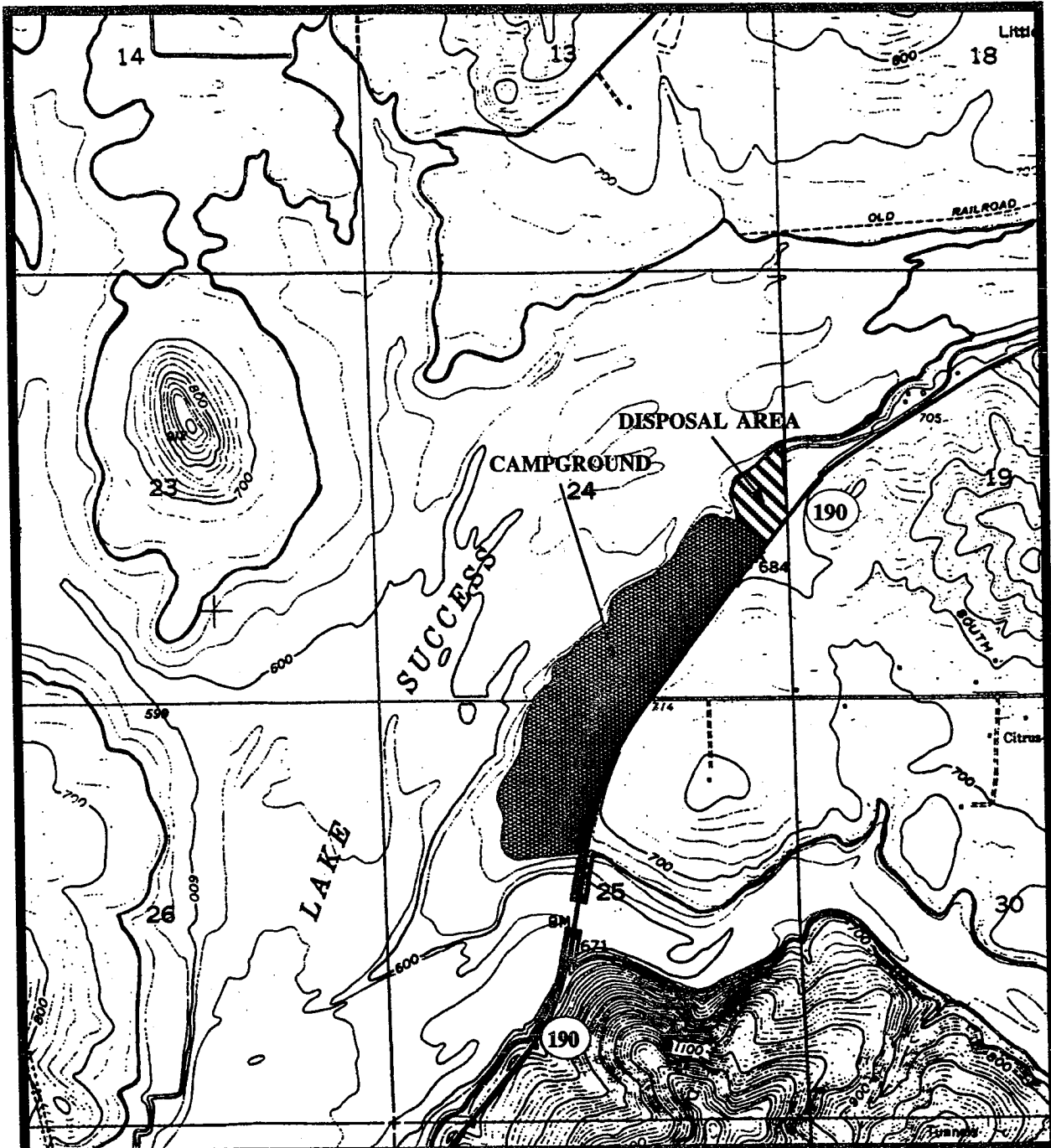
Ordered by:


GARY M. CARLTON, Executive Officer

11 June 1999

(Date)

SH:fmc:6/11/99



ATTACHMENT A

Vicinity Map

U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

Sections 24 & 25, T21S, R28E, MDB&M
Success, 7 1/2' USGS Quad, 1956

INFORMATION SHEET

ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY

U.S. Army Corps of Engineers owns and operates a campground at Lake Success. The campground is between Lake Success and Highway 190, approximately two miles northeast of Success Dam and seven miles east of Porterville.

The facility consists of a 104-unit campground, which includes seven restrooms and a recreation vehicle pump out station. Treatment facilities for the domestic waste consist of five septic tanks. Effluent from the septic tanks is transported via force main to the disposal area. The disposal area consists of two evaporation/percolation ponds occupying approximately one acre of land. Only one pond is regularly used with the second pond serving as a backup. The second pond has not been used since construction. The pond area is removed from the campsite and is completely enclosed by a 6-foot cyclone fence.

Soils at the disposal area consist of San Joaquin Loam. The surface layer of this soil is typically brown to reddish brown loam, about 13 inches thick. The subsoil is reddish brown sandy clay loam and clay about 12 inches thick. Below the subsoil is a yellowish red hardpan about 31 inches thick. Below the hardpan is brown, stratified sandy loam and loam to a depth of 78 inches. This San Joaquin soil has very slow permeability in the subsoil and in the hardpan.

The average annual rainfall in the area is 11 inches and annual average pan evaporation rate is 82 inches.

The Order has a EC limit of source water EC plus 500 $\mu\text{mhos/cm}$. This is based on the maximum allowable limit in the Water Quality Control Plan for the Tulare Lake Basin, Second Edition (Basin Plan). The treatment facility receives wastes at its dump station that may cause the effluent to approach the maximum allowable limit.

Wastewater disposal ponds have the potential to create odor nuisance conditions. Therefore, the Order requires that objectionable odors from the ponds not be perceived beyond the disposal area. Beyond this area the public is present and might be affected. When dissolved oxygen is low, the wastewater could become septic and cause bad odors. Therefore, the Order requires that the dissolved oxygen of the ponds be greater than 1 mg/l to eliminate potential of bad odors to be generated at the ponds.

To operate without problems, wastewater disposal ponds should have near neutral pH and normal algae growth. A pH less than 6.5 indicates acidic conditions and greater than 8.5 indicates basic conditions, too extreme for effective operation. The Order requires that the ponded water have pH between 6.5 and 8.5.

**INFORMATION SHEET - ORDER NO. 99-081
U.S. ARMY CORPS OF ENGINEERS ORDER NO.
SUCCESS RESERVOIR CAMPGROUND
TULARE COUNTY**

-2-

Ponded water creates a potential for mosquitoes to breed and create vector nuisances. The Order has requirements on pond maintenance such that this threat is minimized. Also, in order to prevent overflow conditions or failure of bank embankments, the Order requires the ponds to have adequate capacity with a minimum of two feet of freeboard.

This Order includes groundwater limitations based on the Basin Plan.

This Order requires the Discharger to either pump the septic tanks at least annually or inspect it annually to make sure that the septic system operates effectively so that neither solids nor grease are carried over to the ponds. To assure that pumped septage from the septic tanks is properly handled and dumped at an authorized place for effective treatment, the Order requires the Discharger to use only licensed septic tank cleaning services.

Chemicals used for control of odors in RV tanks usually contain formaldehyde, zinc, and phenols as active ingredients. These chemicals are toxic at certain concentrations. The Order includes narrative language that requires the discharge to be such that these chemical constituents will not adversely affect the treatment system and affect groundwater. Also, the Discharger is required to monitor these constituents annually.

Domestic wastes contain nitrogen. Nitrogen in nitrate form is readily leachable. Therefore, it has the potential to affect groundwater. The Order requires the Discharger to monitor the effluent for nitrogen.

This is an existing facility and the action to update requirements is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, Section 15301.

SH:fmc:6/11/99